

Sociogenomics

Mendelian Randomization

Nicola Barban

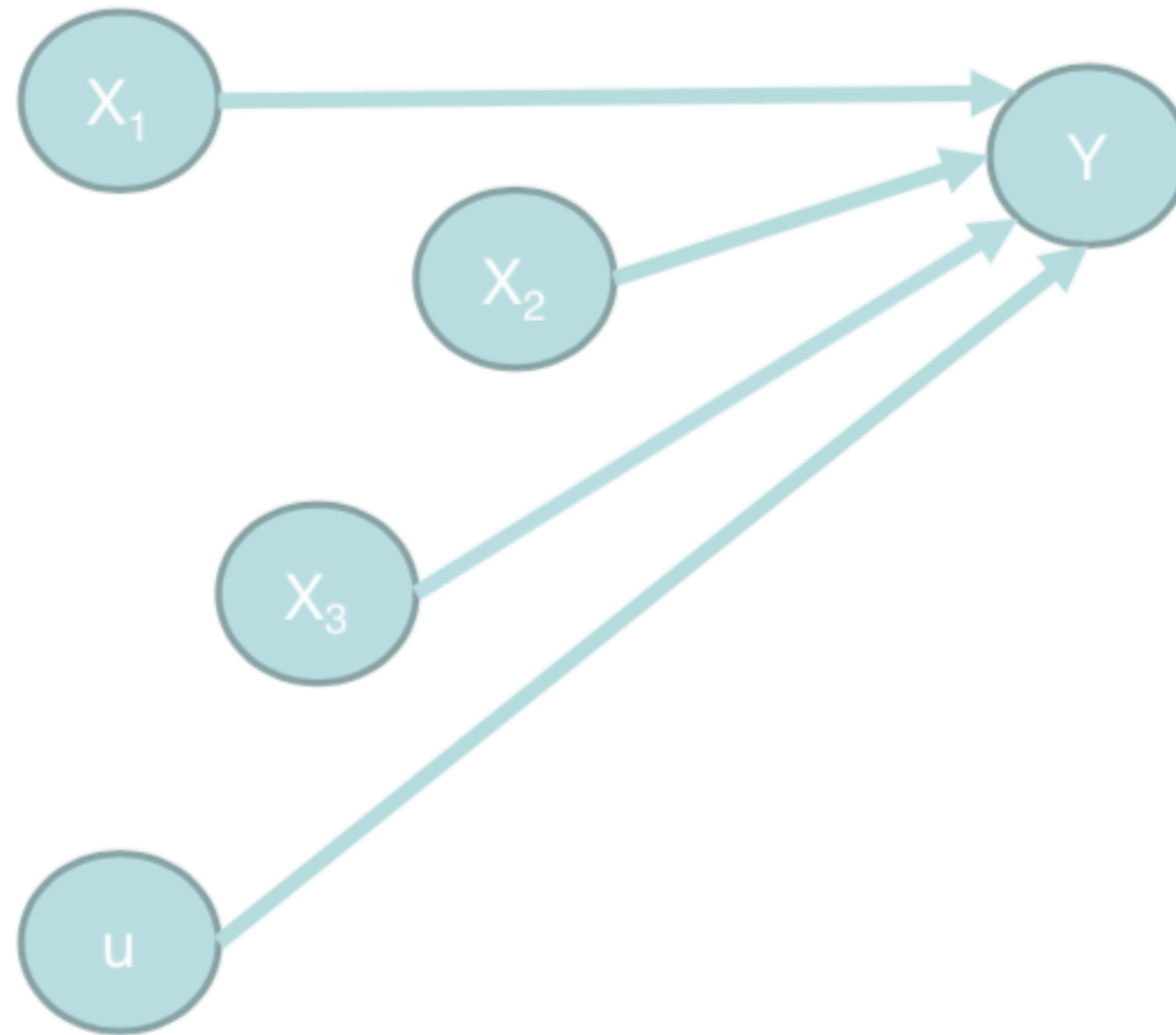


How we study causal effect of X on Y?

- Regression (Ordinary Least Squares estimation)
- $Y_i = \alpha + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i} + u_i$

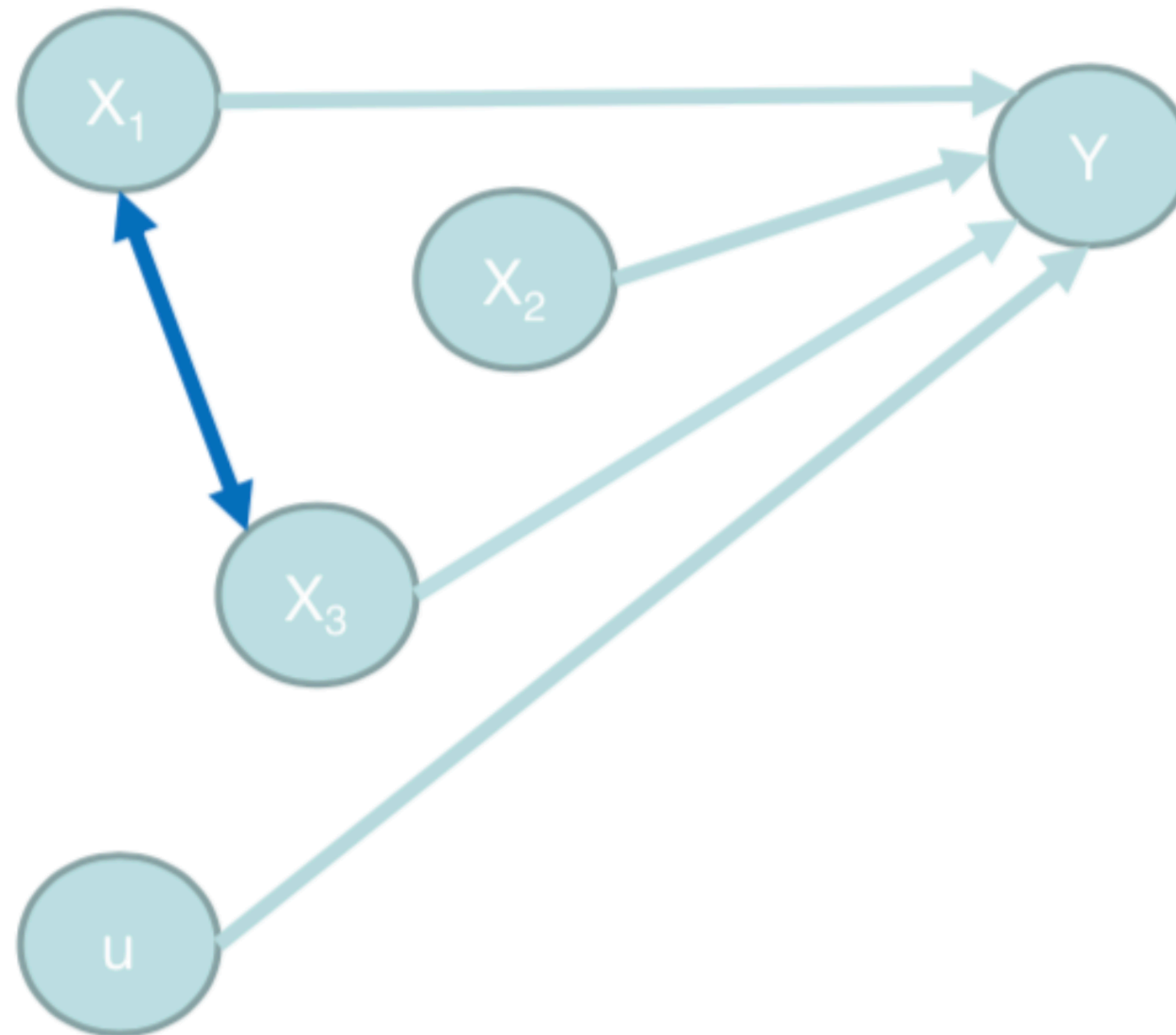
Standard regression

- No association between X_1 and u
- OLS estimates are consistent



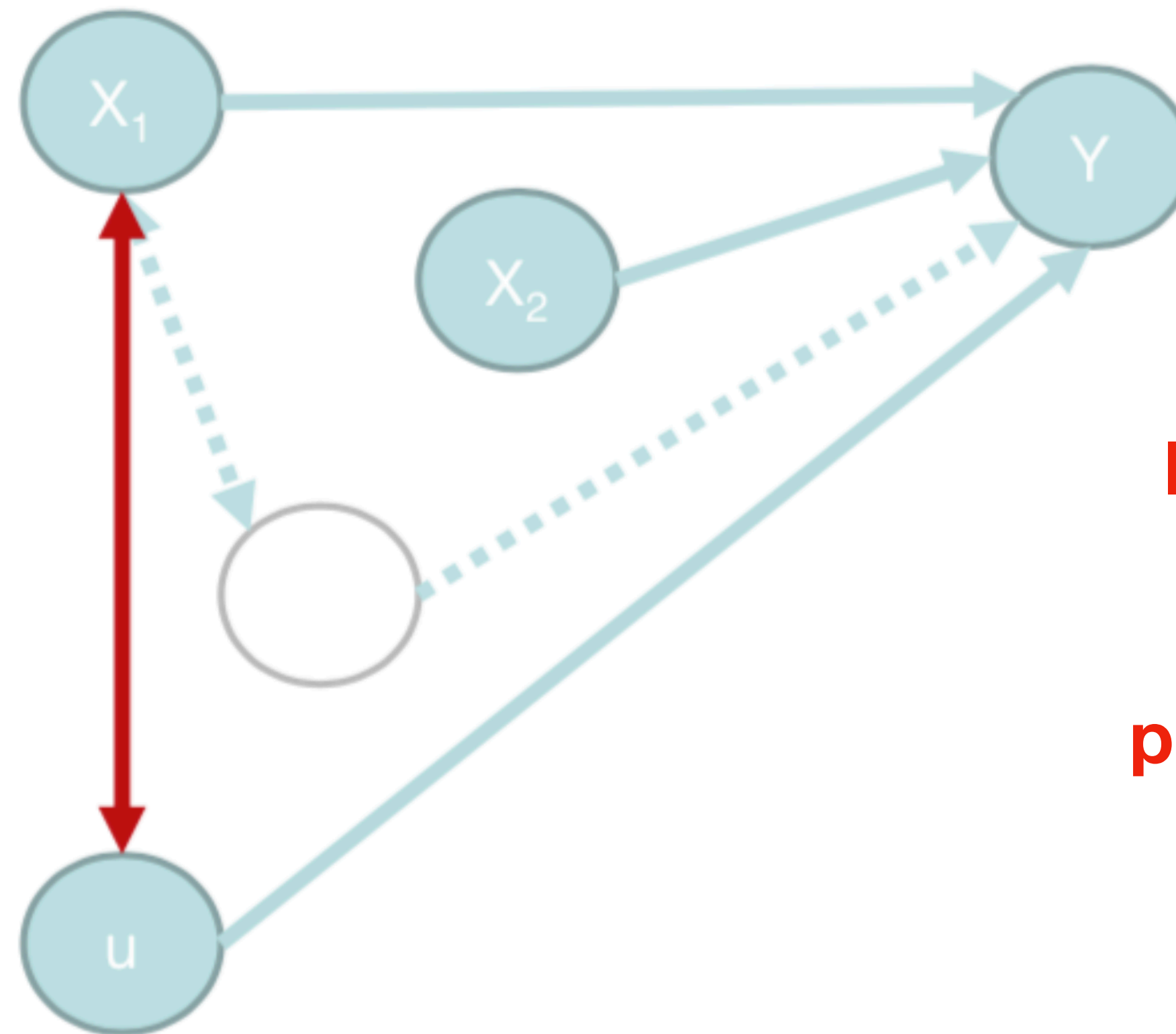
Standard regression

- No association between X_1 and u
- OLS estimates are consistent, even if association between X_1 and X_3



Endogeneity

- Association between X_1 and u if X_3 is unobserved
- OLS inconsistent



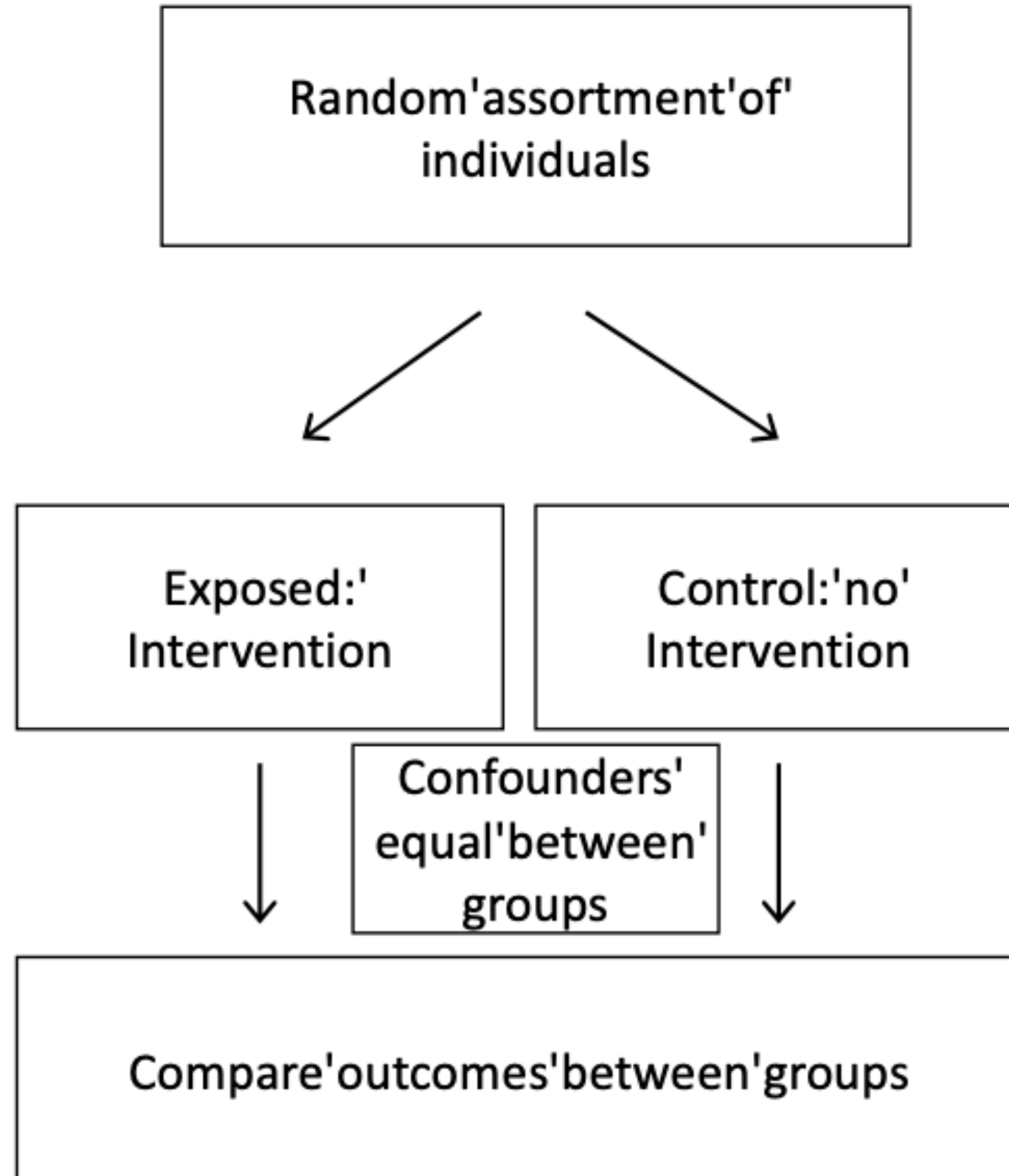
**Inconsistent means
that does not
converge in
probability to the true
value of the
parameter**

Endogeneity

In a statistical model, an **endogenous** variable is one that is correlated with the error term $E[X_j\epsilon] \neq 0$. There are different reasons why this can happen:

- 1 **Omitted variable:** there is a spurious variable that explains both Y and X and has not been included in the regression. However, sometimes this cannot be measured.
- 2 **Measurement error:** there is some non-random measurement error in your variable of interest. In a study of sexual behaviour, for example men overreport the number of sexual partners, women underreport. Or high earners underreport their earnings and wealth.
- 3 **Reverse causation:** sometimes is not that X causes Y , but Y causing X . If we study the effect of schooling on fertility, we cannot exclude that teenage parents drop school because pregnant.

Randomized control trial



Hierarchy of evidence

When the goal is to infer causation...

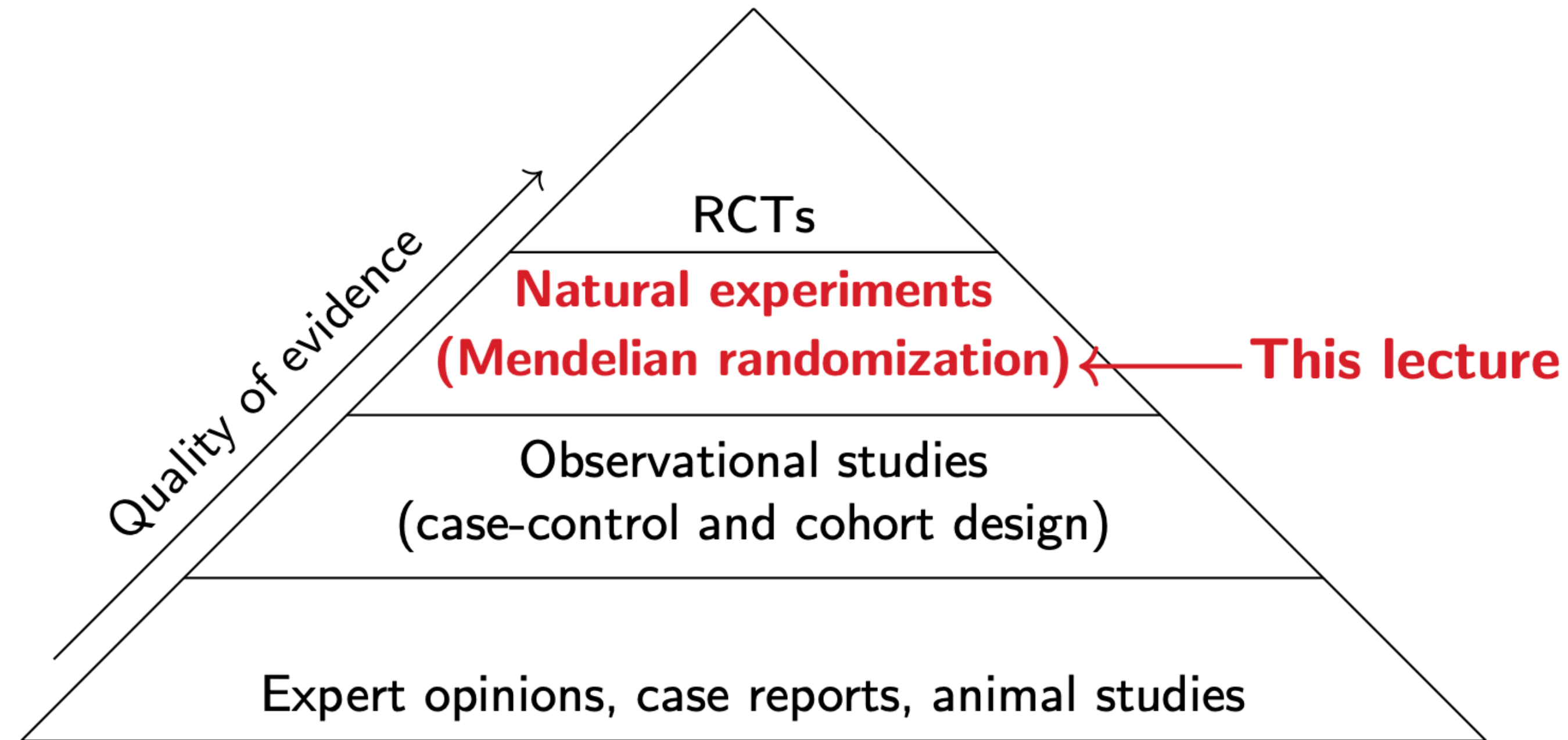


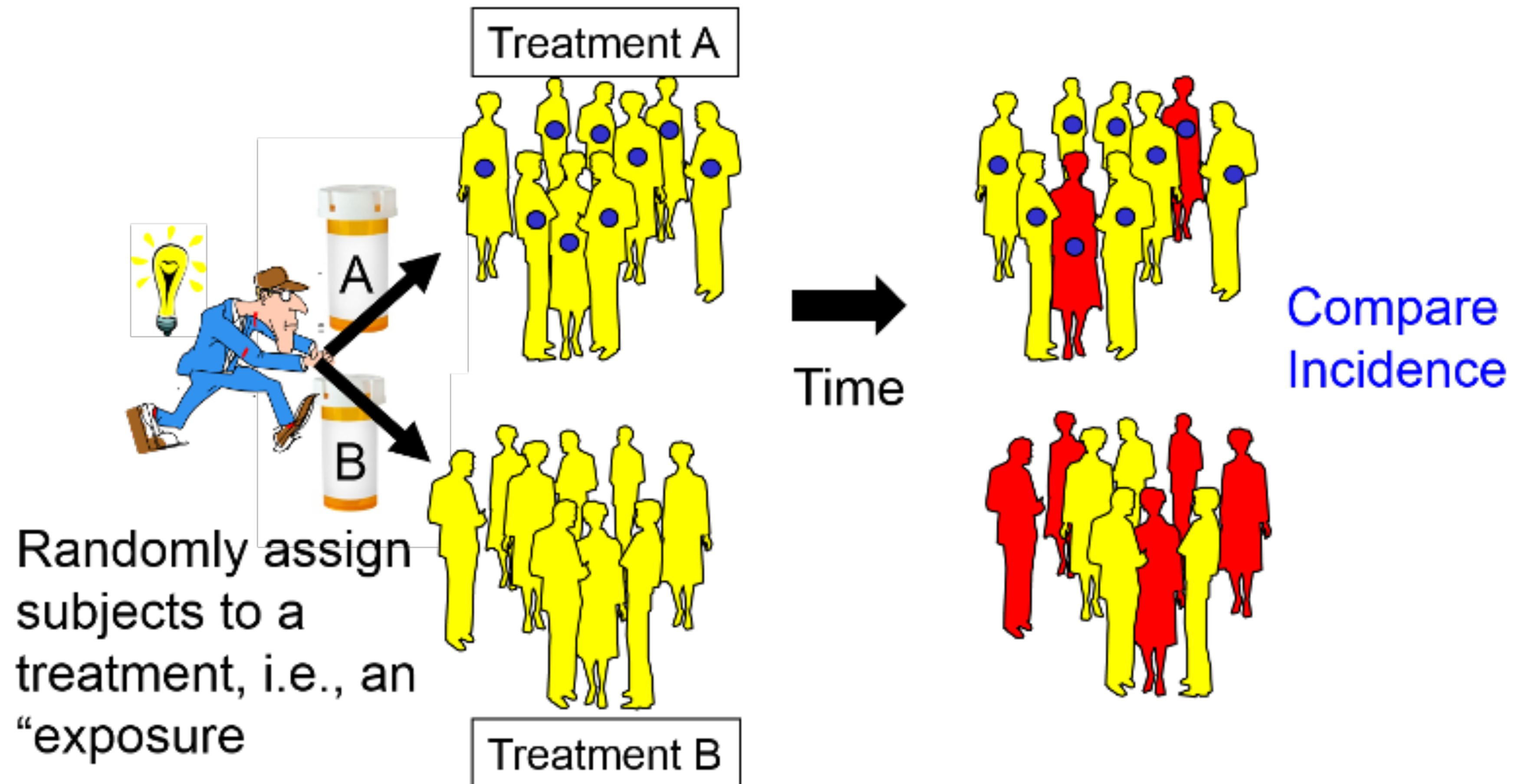
Figure: (A rough) Hierarchy of evidence in medical studies.¹

¹Based on: American Academy of Pediatrics clinical guidelines. Gidding, et al. (2012). "Developing the 2011 Integrated Pediatric Guidelines for Cardiovascular Risk Reduction." *Pediatrics* 129(5).

RCT

Randomized control Trials

- Not always possible
- Sometimes:
 - Not feasible
 - Not ethical
 - Data not available



Correlation

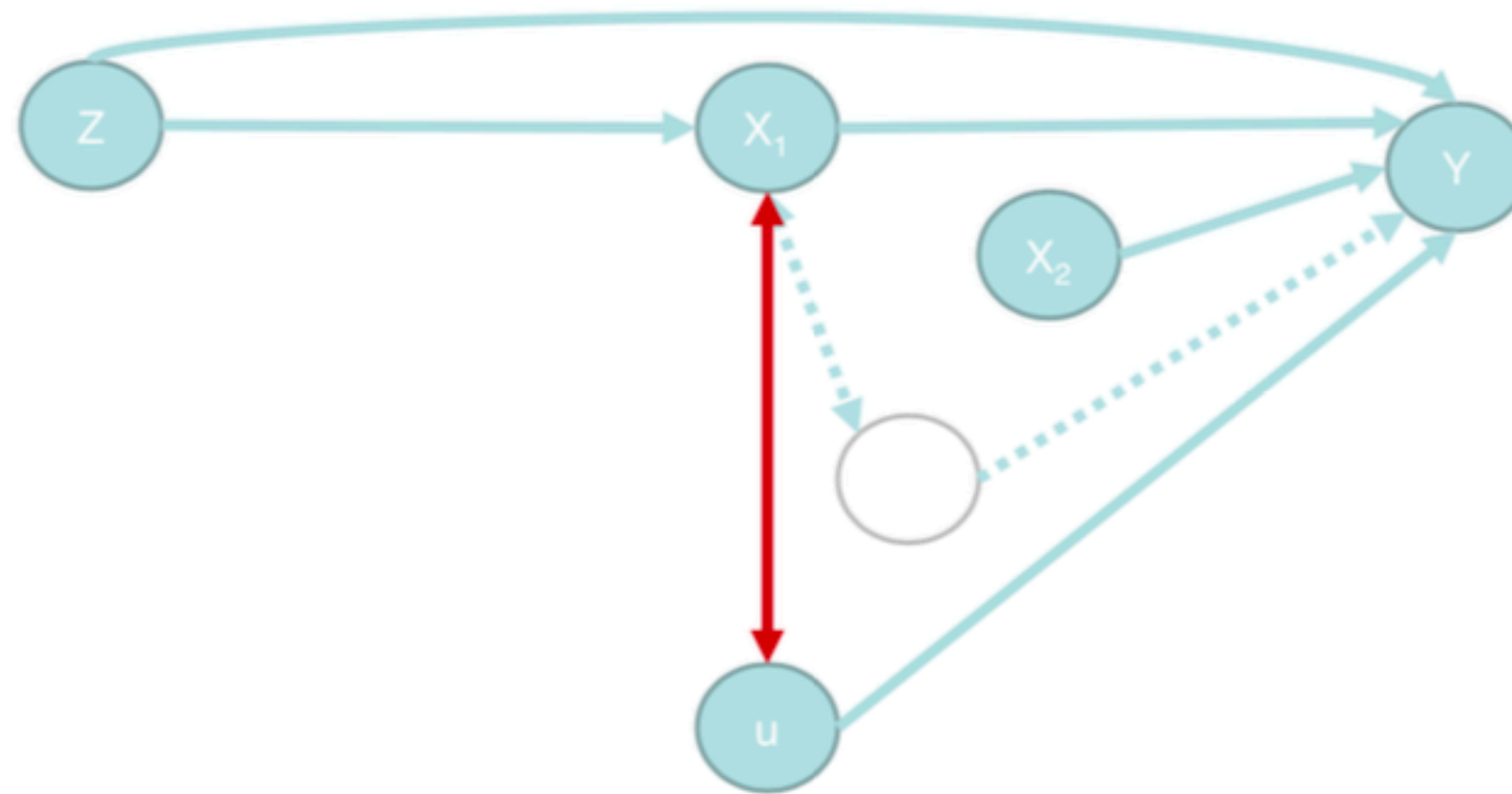


Causality

<https://www.tylervigen.com/spurious-correlations>

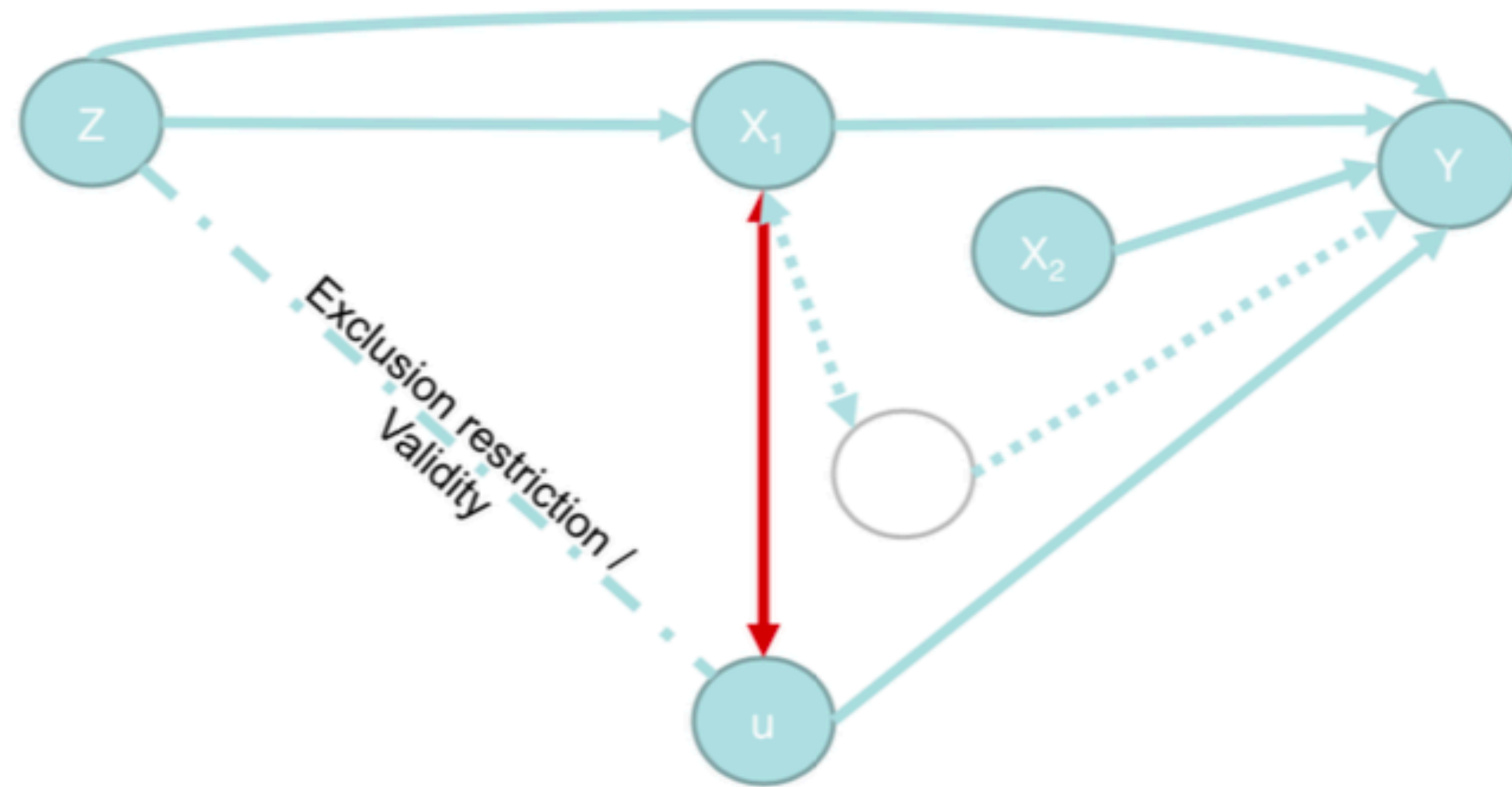
Instrumental variable

- Z must be associated with X_1



Instrumental variable

- Z must be associated with X_1
- Z must not be associated with u



Example of IV:

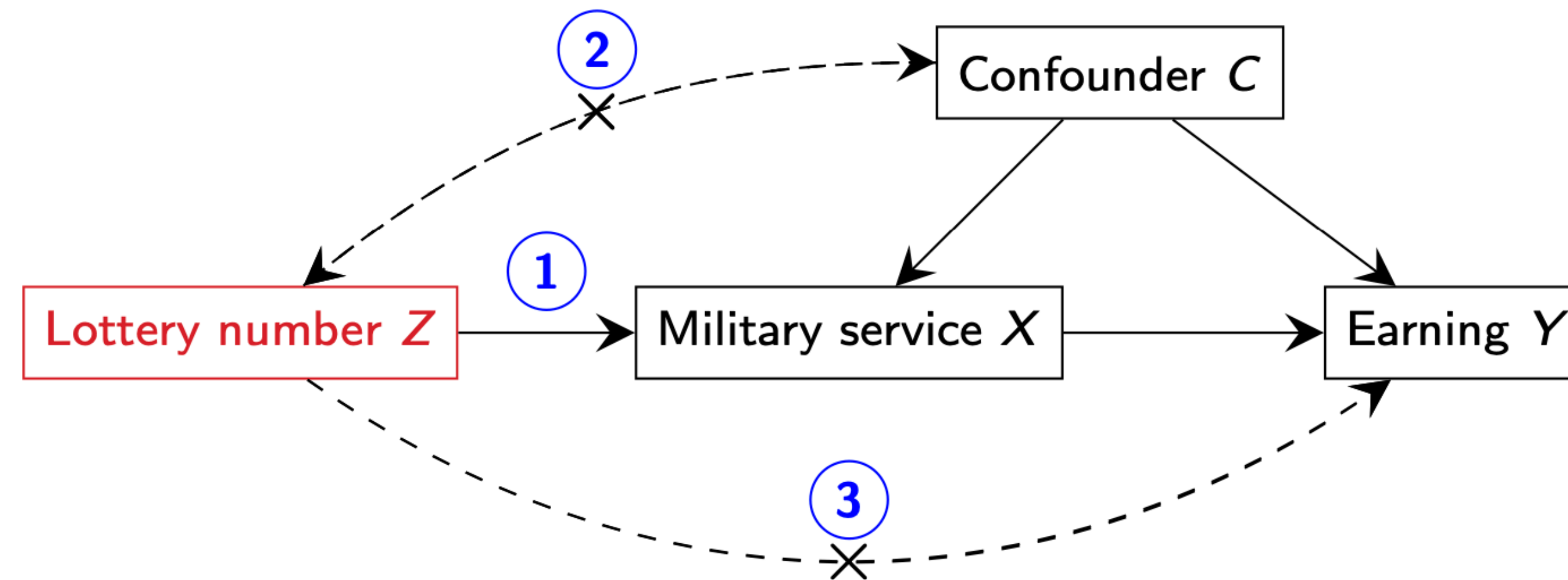
Angrist, Joshua. 1990. "Lifetime Earnings and the Vietnam Era Draft Lottery." *American Economic Review* 80: 313-336.

Does military service cause changes in earnings?

- Unobserved factors might affect both military service and earnings.
- Televised drawing of Random Sequence Numbers (1-365) which assigned draft priorities to birth dates
- Men were drafted in 1970-1972
- the results agree with the claim that military service reduces earnings through lost work experience.



IV in Economics: Effect of military service on earnings²



- The draft lottery can be regarded as a “natural experiment” of military service.

²Angrist, J. (1990). Lifetime earnings and the Vietnam era draft lottery: evidence from social security administrative records. *American Economic Review*, 80(3), 313–336.

Instrumental variables

- A variable that is correlated with X but not with Y
- The best analogy is to think to a lottery

The genetic lottery

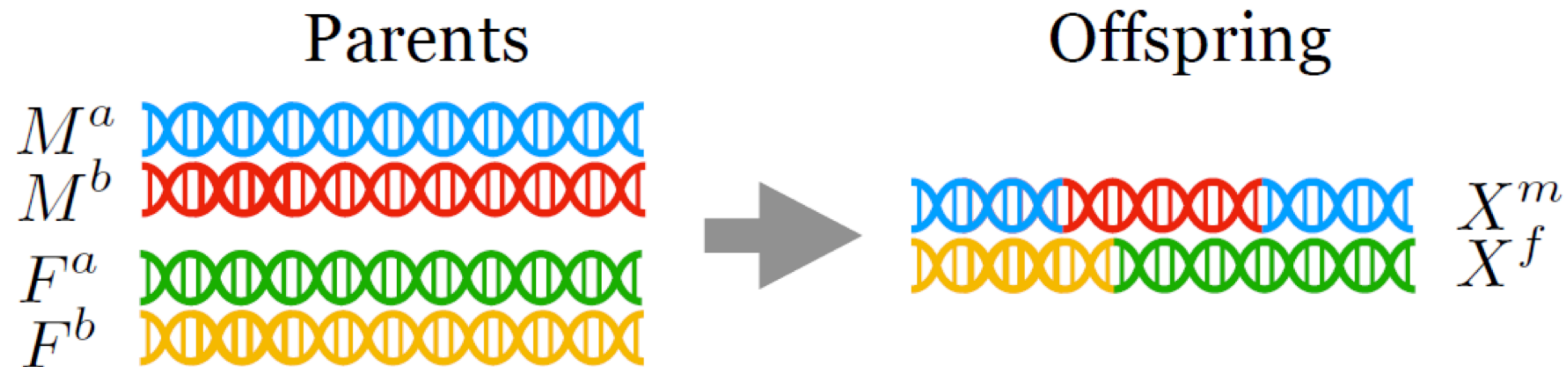
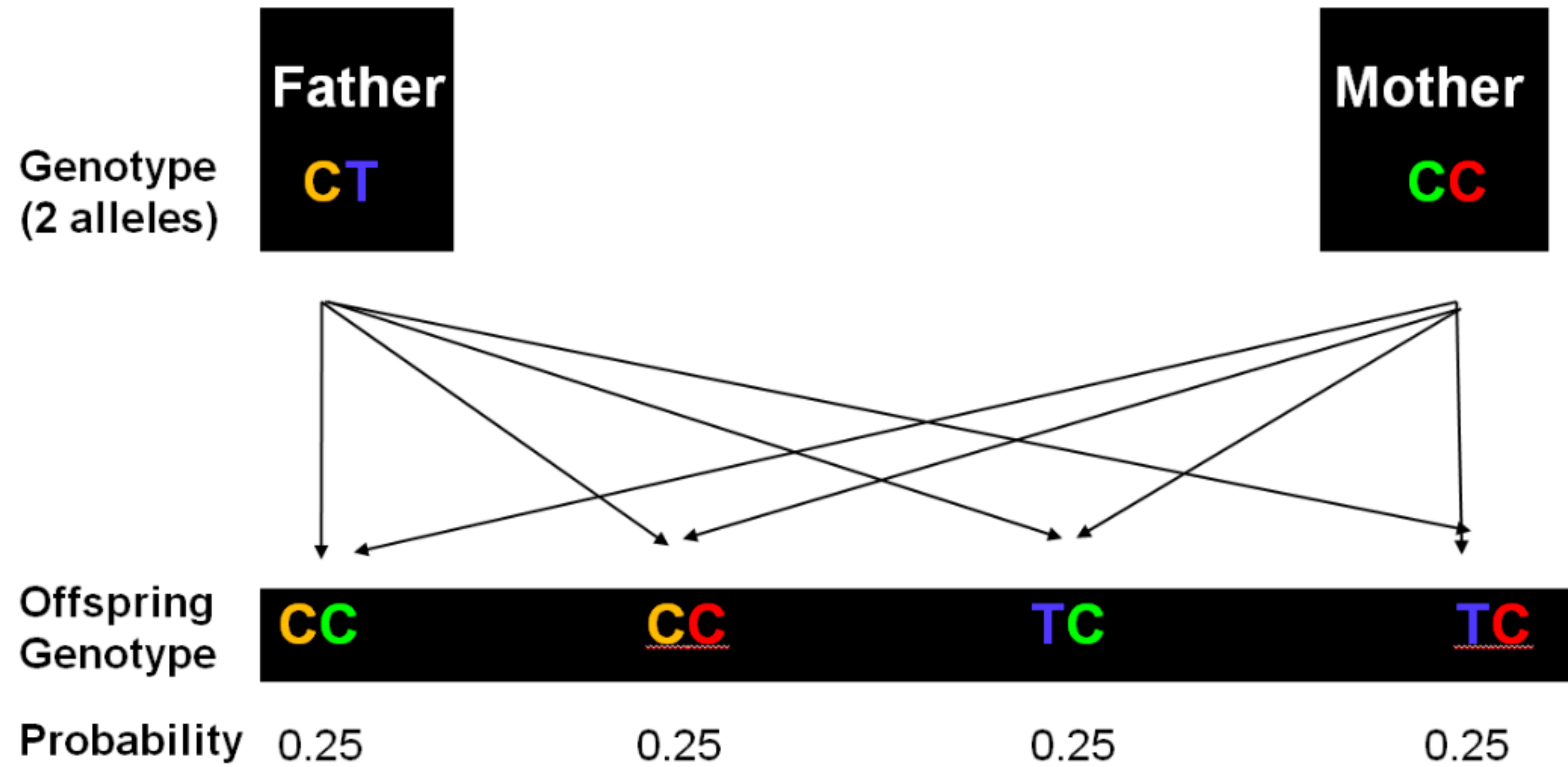
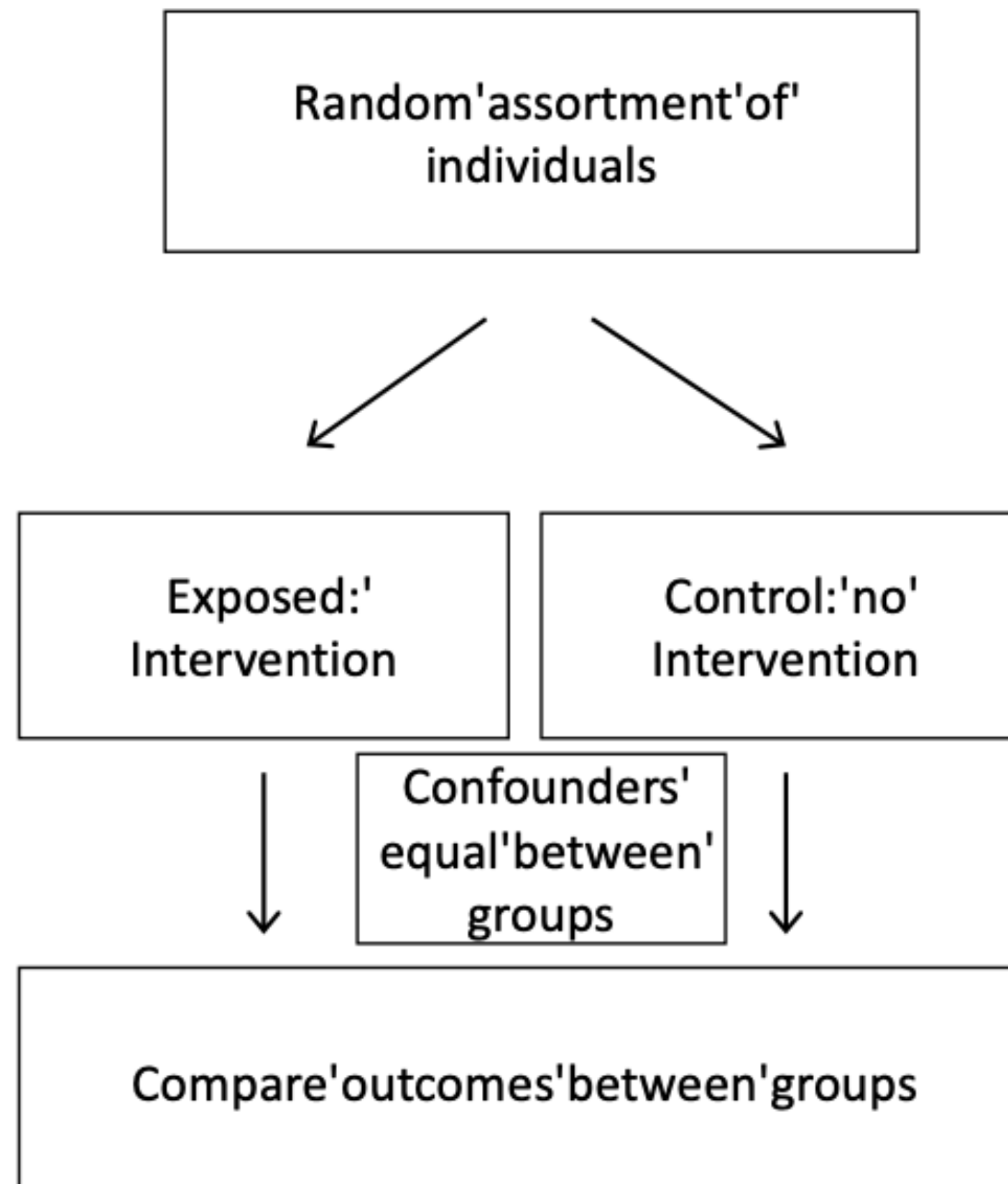


Figure from Bates *et al.* Causal Inference in Genetic Trio Studies,
PNAS, 2021

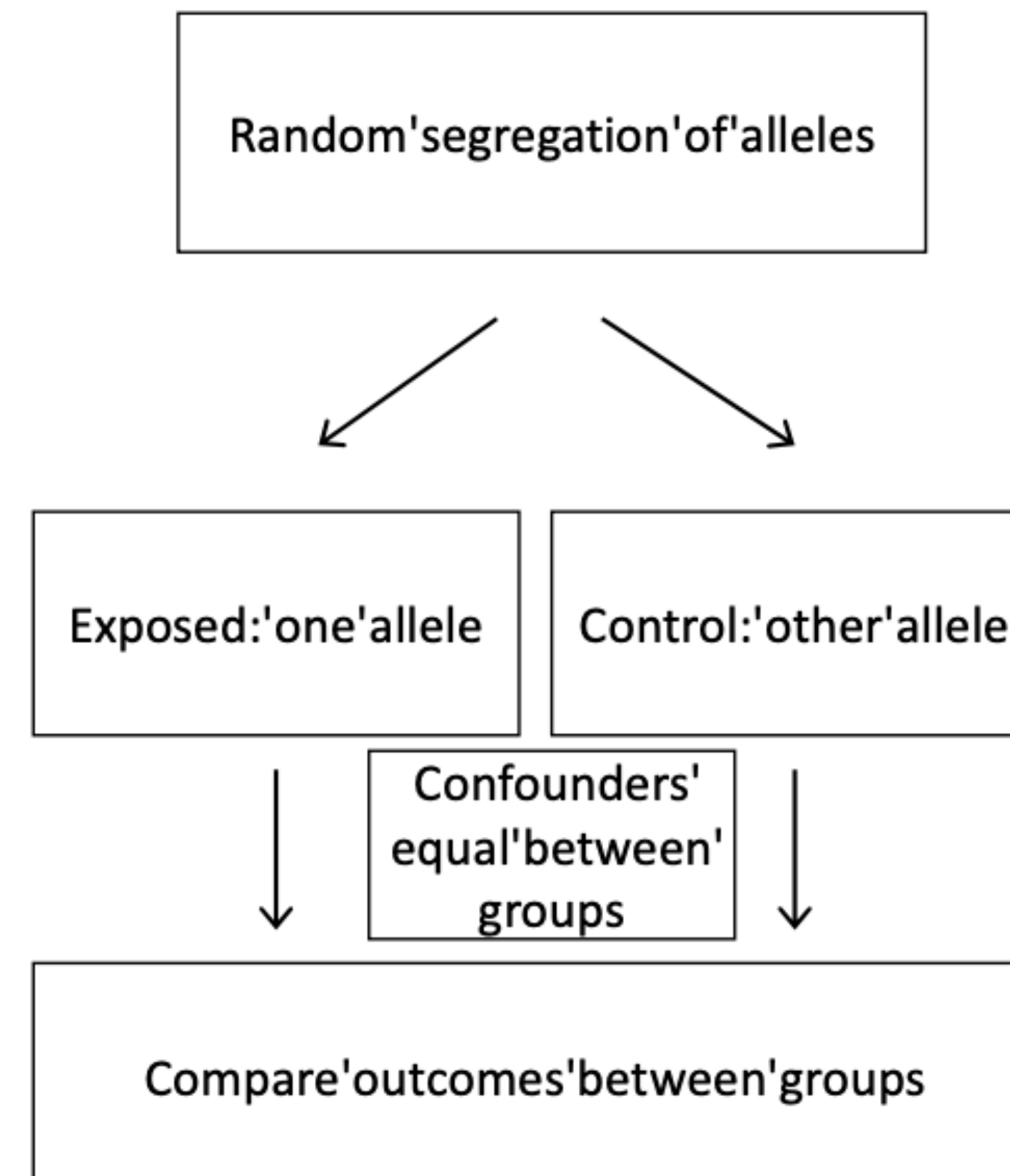
Equal chance of passing 1 of 2 alleles with each conception

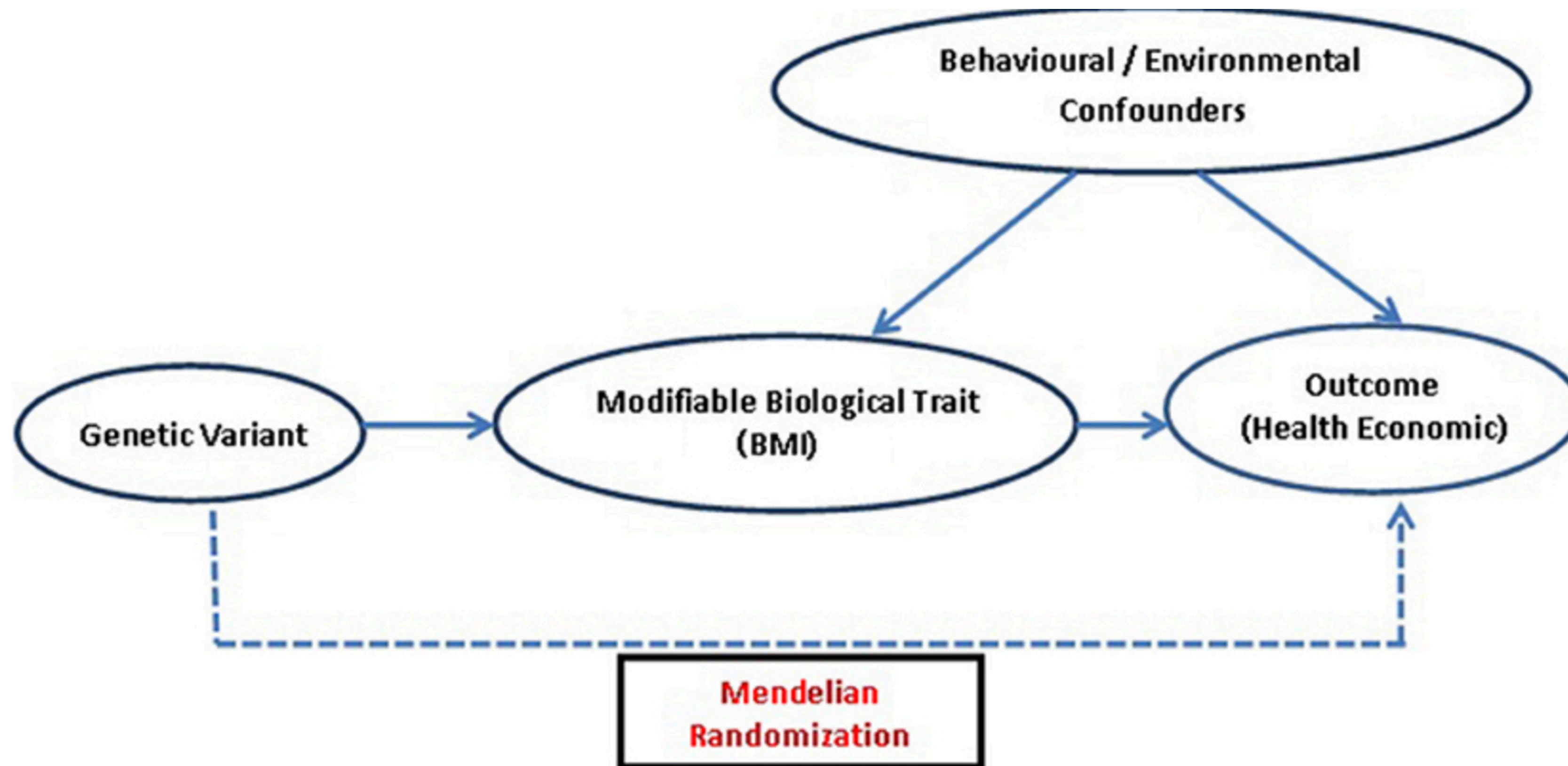


Randomized control trial



Mendelian Randomization

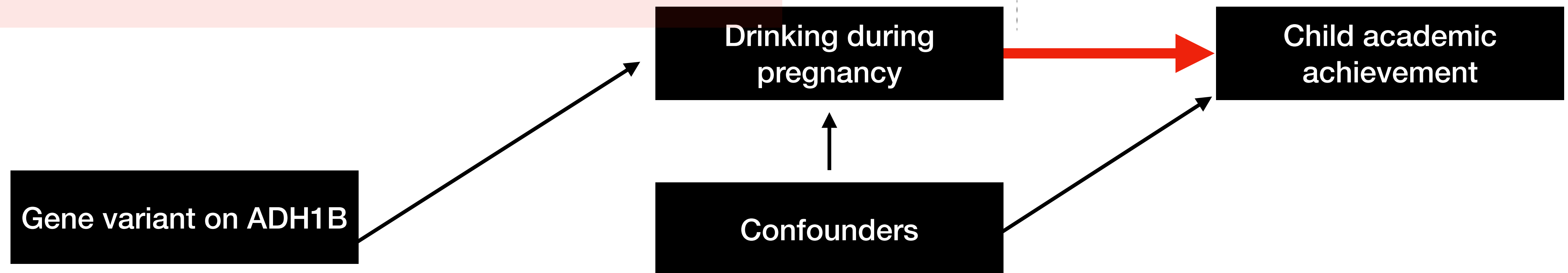




ALCOHOL EXPOSURE *IN UTERO* AND CHILD ACADEMIC ACHIEVEMENT*

Stephanie von Hinke Kessler Scholder, George L. Wehby, Sarah Lewis and Luisa Zuccolo

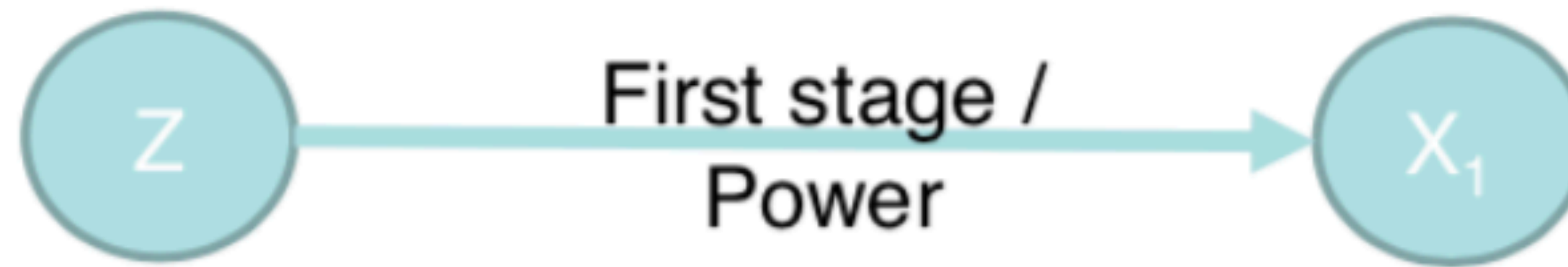
We examine the effect of prenatal alcohol exposure on child academic achievement. We use a genetic variant in the maternal alcohol-metabolism gene *ADH1B* to instrument for alcohol exposure, whilst controlling for the child's genotype on the same variant. We show that the instrument is unrelated to an extensive range of parental characteristics and behaviour. OLS regressions suggest an ambiguous association between alcohol exposure and attainment but there is a strong social gradient in drinking, with mothers in higher socio-economic groups more likely to drink. In contrast to the OLS, the IV estimates show clear negative effects of prenatal alcohol exposure.



Two Stages Least Squares 2SLS

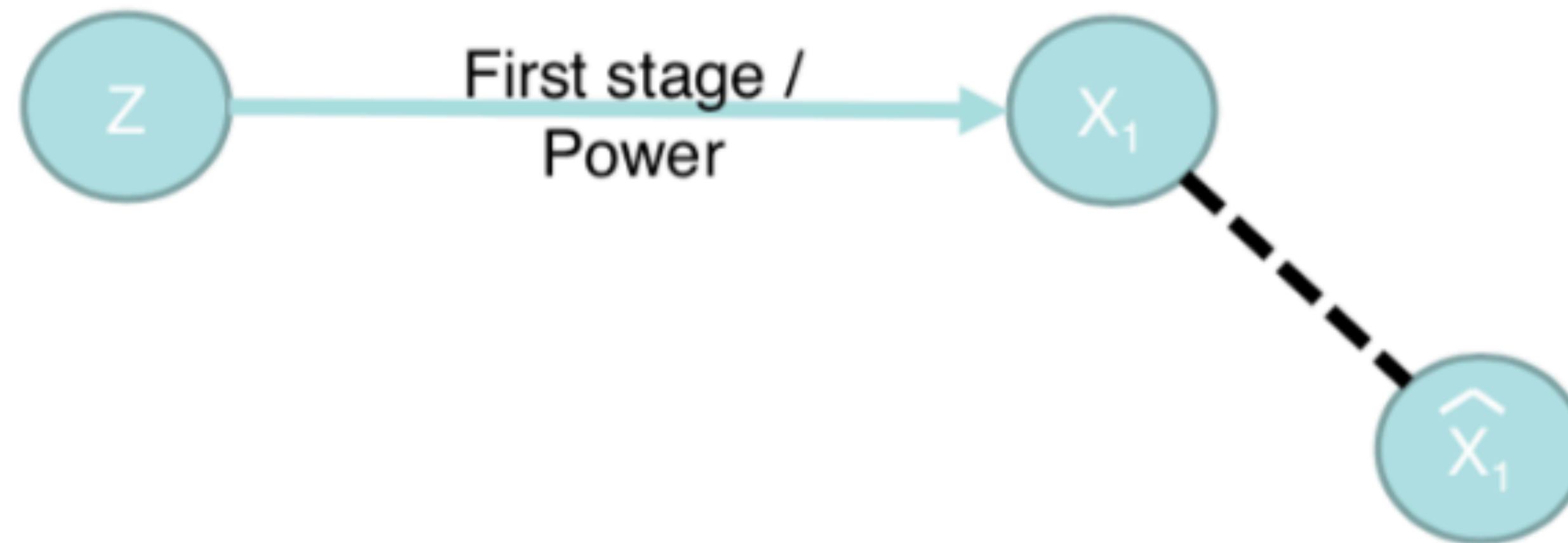
First stage

- Regress (endogenous X_1) on Z



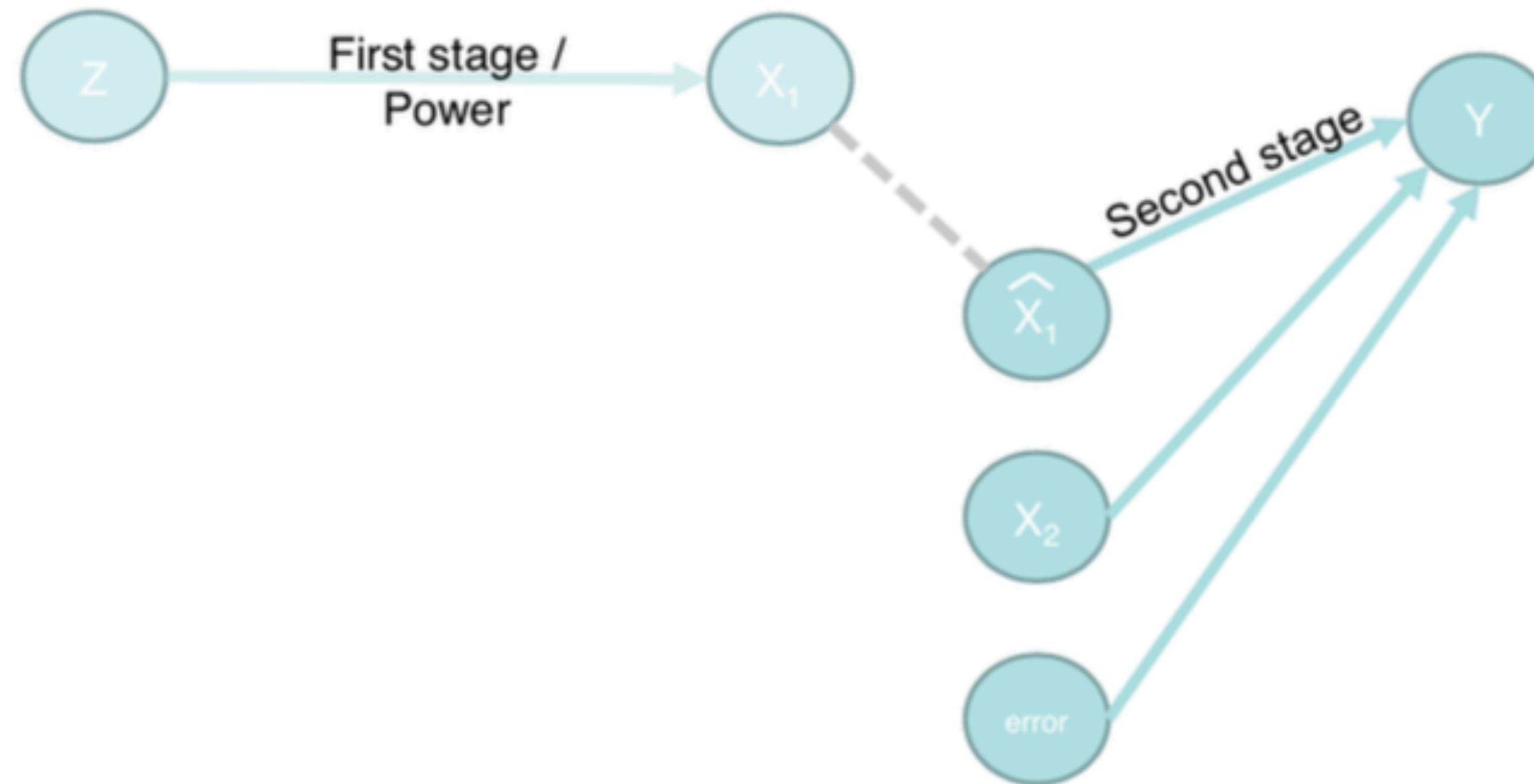
First stage

- Regress (endogenous X_1) on Z

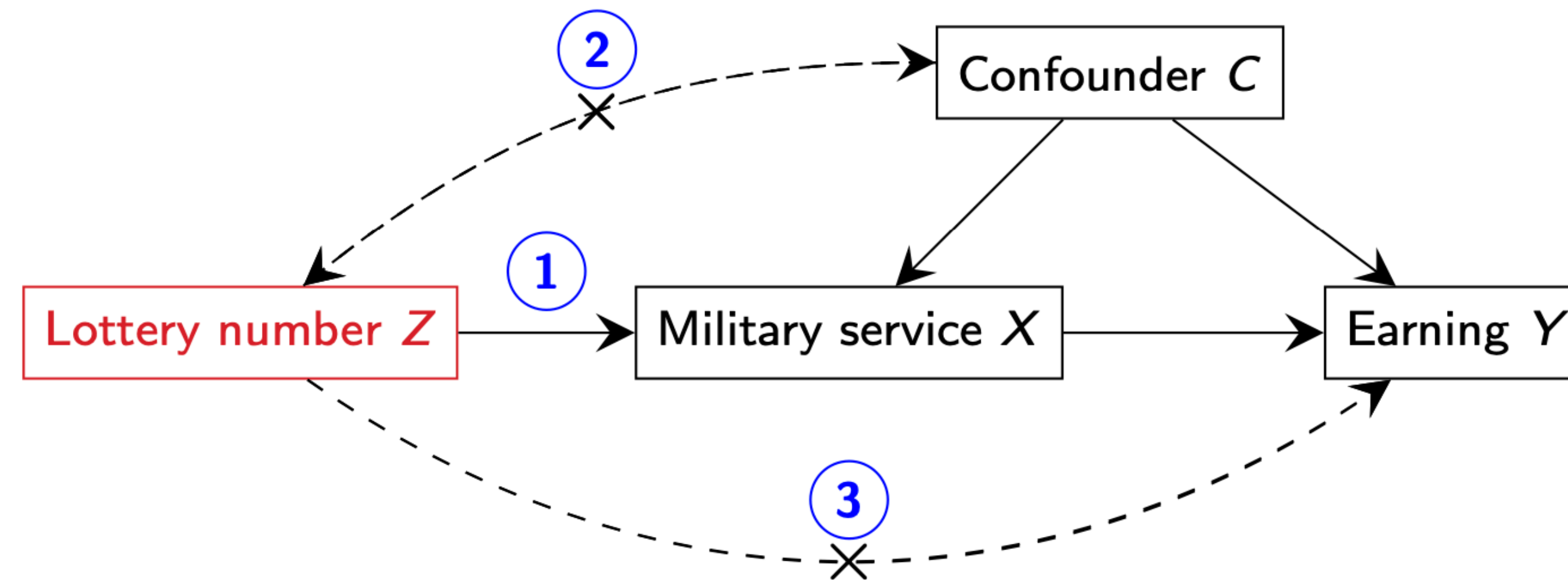


Second stage

- Use the predicted values of X_1 to estimate Y



IV in Economics: Effect of military service on earnings²



- The draft lottery can be regarded as a “natural experiment” of military service.

²Angrist, J. (1990). Lifetime earnings and the Vietnam era draft lottery: evidence from social security administrative records. *American Economic Review*, 80(3), 313–336.

IV glossary

- First-Stage regression:

$$X_i = \alpha_1 + \phi Z_i + e_{1i}$$

- Second-Stage regression:

$$Y_i = \alpha_2 + \lambda \hat{X}_i + e_{2i}$$

- Reduced form (links instrument and outcomes):

$$Y_i = \alpha_0 + \rho Z_i + e_{0i}$$

Two Stage Least Squares

The two stage least square is a general way to estimates instrumental variables when we control for covariates

- Reduced form: $Y_i = \alpha_0 + \rho Z_i + \gamma_0 A_i + e_{0i}$
- First stage: $D_i = \alpha_1 + \phi Z_i + \gamma_1 A_i + e_{1i}$
- The 2SLS: $Y_i = \alpha_0 + \lambda_{2SLS} \hat{D}_i + \gamma_2 A_i + e_{2i}$

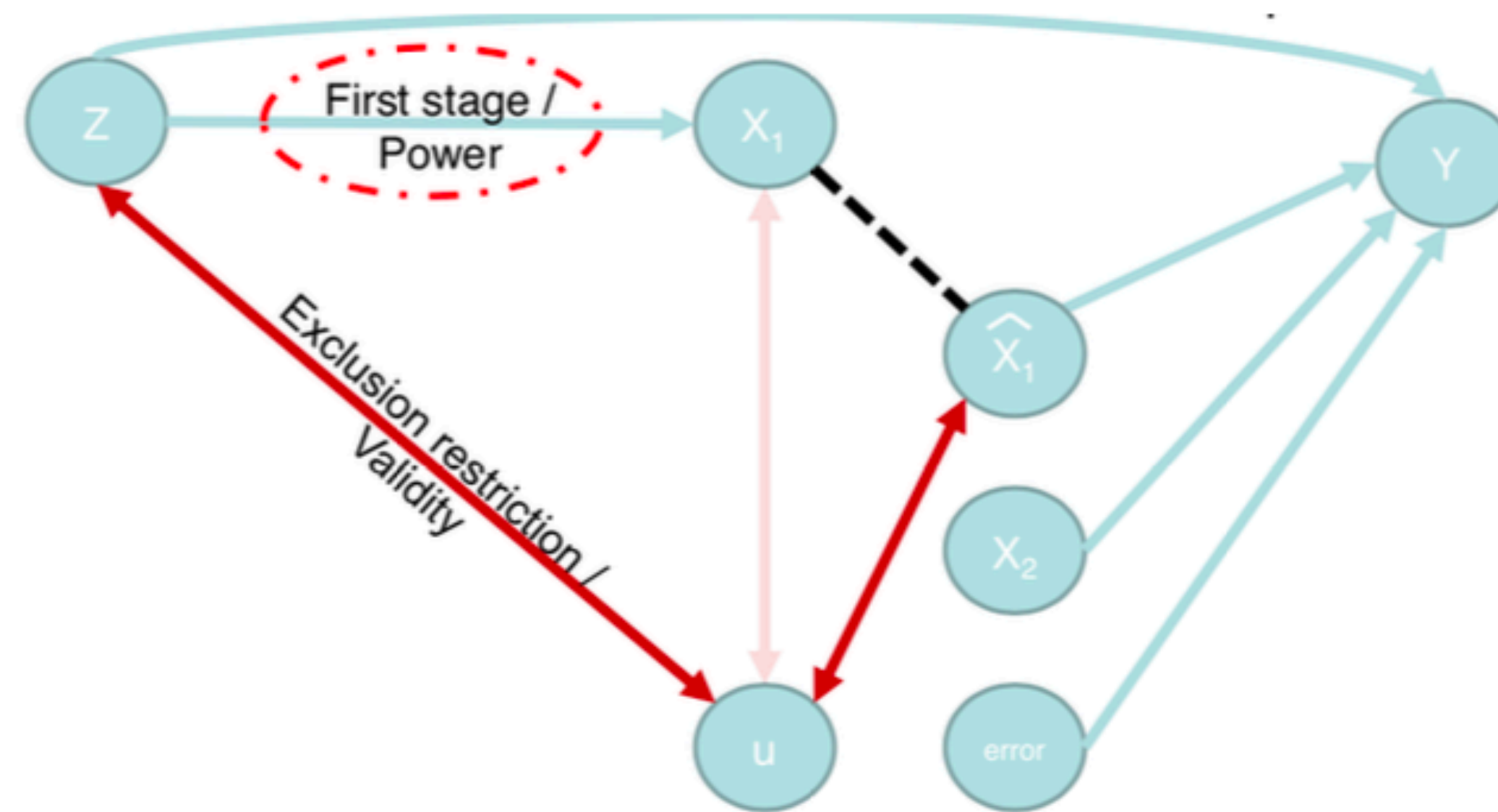
In this case, these coefficient are estimated with A_i included as covariates.

$$\lambda_{2SLS} = \frac{\rho}{\phi} = \frac{COV(Y_i, \tilde{Z}_i)/V(\tilde{Z}_i)}{COV(X_i, \tilde{Z}_i)/V(\tilde{Z}_i)},$$

where $\tilde{Z}_i$ is the residual from a regression of Z_i on A_i (remember regression anatomy)

Two critical assumptions for the validity of an instrument

- 1 There must be no association between Z and u (**instrument exogeneity or exclusion restriction**)
- 2 There must be sufficient correlation between Z and X_1 (**instrument relevance**)



Assumption I: Relevance

- There must be sufficient correlation between Z and X
- **This can be proven using the data**
- When the instruments are not relevant, they are called **weak instruments**.
- Rule of thumb: the first-stage F-statistic testing the hypothesis that the coefficient on the instrument is zero should be at least 10 (for a single endogenous regressor)
- A small F-statistic means the instrument is weak (it explains little of the variation in X) and the estimator is biased

Assumption II: exclusion restriction

- There must be no association between Z and u
- This cannot be demonstrated using the data
- Researchers must appeal to existing theory and evidence. You have to tell a convincing story about why a particular IV is a good instrument!!
- Something to consider when thinking about whether a particular IV is ‘good’
- → Does the IV, for all intents and purposes, randomize the endogenous regressor?

1st stage in mendelian randomisation

- Association between GENES and X can be assessed through genetic association and GWAS
- Single SNP effect can affect power of 1st stage —> **weak instrument**
- **Multiple SNP effects can be aggregated through scores (PGS)**
-

Exclusion restriction in Mendelian Randomization

- Genes need to have a unique effect on X and not directly associated with outcome
- **PROBLEM:**
 - **Linkage disequilibrium.** Causal genes may be in disequilibrium with variants associated with outcomes
 - **Pleiotropy,** genes associated with X may be associated with another outcome that influences Y

Other assumptions

TABLE 13.1 Summary of the main assumptions in MR research

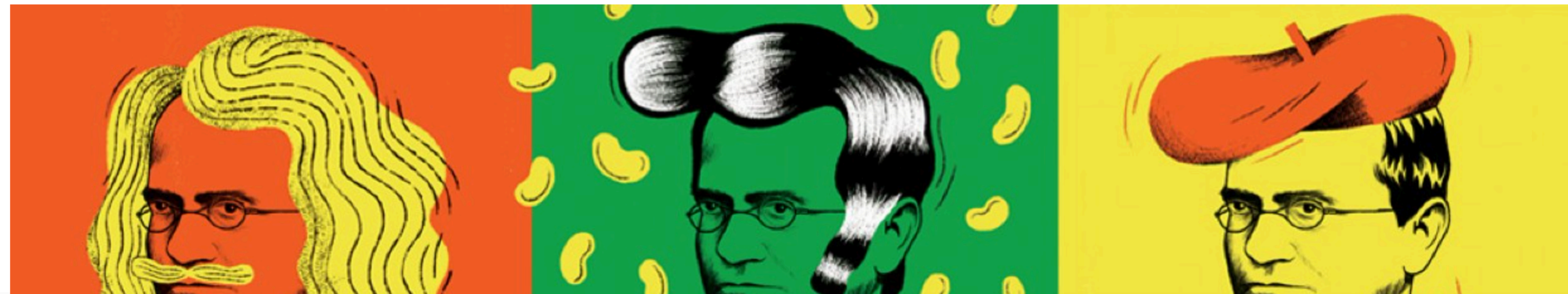
Assumption	Description	Solutions	Limitations
Weak instrument	IV needs to be correlated with the treatment (X) variable If not, can lead to noise and biased estimates	Use PGSs Test strength of association of PGS in independent samples	Lose knowledge about the instruments and potentially introduce pleiotropy
Exclusion Restriction	association between the IV and outcome (Y) variables is entirely due to a causal effect of the exposure on the outcome	Modelling pleiotropic effects, detecting and excluding pleiotropic SNPs, controlling for direct genetic effects on Y independent of X	Quantitative solutions always inherit the shortcomings of observational studies (e. g. measurement error)
Independence assumption	IV and Y do not have a common cause	Control for population stratification	Control might be imperfect, see also Chapter 3
No genetic assortative mating	Partners are not mating based on the same genotypes for X or across X and Y	Controlling for parental genotype	Measurement error
Monotonicity assumption	Genes influence the exposure in the same direction for all individuals	Exclude SNPs with non-monotonic effects	Difficult to identify SNPs with non-monotonic effects due to low power

[nature](#) > [news feature](#) > article

NEWS FEATURE | 10 December 2019

The gene-based hack that is revolutionizing epidemiology

Mendelian randomization offers a simple way to distinguish causation from correlation. But are scientists overusing it?

[David Adam](#)

Examples

- Is cholesterol causing cancer?

[nature](#) > [nature communications](#) > [articles](#) > [article](#)

Article | [Open Access](#) | [Published: 27 September 2018](#)

A Mendelian randomization study of the effects of blood lipids on breast cancer risk

[Christoph Nowak](#)  & [Johan Ärnlöv](#)

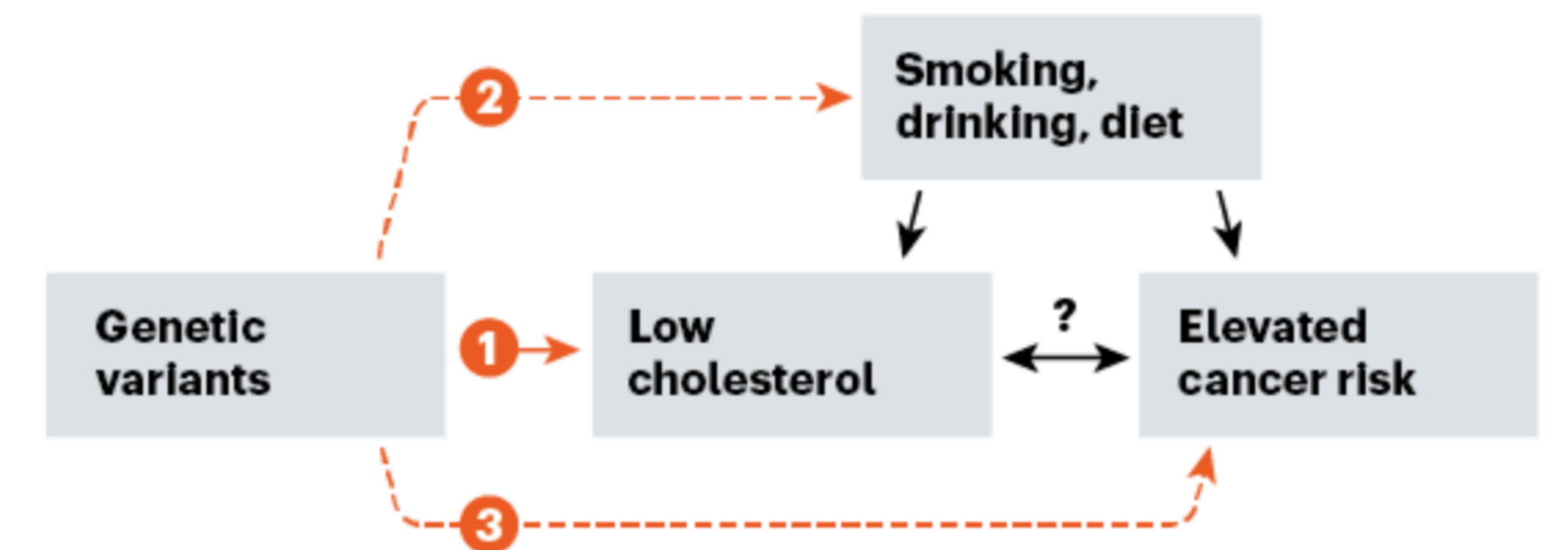
[Nature Communications](#) **9**, Article number: 3957 (2018) | [Cite this article](#)

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Abstract

GENES AS PROXY

Mendelian randomization uses gene variants to interrogate cause and effect in epidemiological data. For example, researchers can ask whether having low cholesterol increases cancer risk — as some data suggest — by looking at people genetically predisposed to low cholesterol. This strategy can help rule out reverse causation (that cancer lowers cholesterol), and it can bypass some variables that might influence both cancer risk and cholesterol. However, it relies on several assumptions, which must be tested.



Cholesterol and breast cancer

- Observational studies have reported inconsistent associations between circulating lipids and breast cancer risk.
- Using Mendelian randomization, we show that genetically raised LDL-cholesterol is associated with higher risk of breast cancer (odds ratio, OR, per standard deviation, 1.09, 95% confidence interval, 1.02–1.18, $P=0.020$) and estrogen receptor (ER)-positive breast cancer (OR 1.14 [1.05–1.24] $P=0.004$).
-

Education and myopia: assessing the direction of causality by mendelian randomisation

- The global prevalence of myopia, or short sight, has risen rapidly in recent decades and has become one of the leading causes of visual disability worldwide
- recent genome wide association studies (GWAS) have identified about 100 genetic variants associated with myopia,
- Myopia has been associated with higher levels of educational attainment for more than a century.
- **unknown whether exposure to more years in education causes myopia;** whether children with myopia might be more intelligent or might spend more time on near work, leading to better educational outcomes; or **whether associations with socioeconomic confounds lead to both myopia and higher levels of education.**

Conclusions This study shows that **exposure to more years in education contributes to the rising prevalence of myopia.** Increasing the length of time spent in education may inadvertently increase the prevalence of myopia and potential future visual disability.

MR and GWAS

- It is now standard to examine causal direction of SNPs on multiple outcomes using MR in when analysing traits with GWAS
- Additional techniques are used to relax assumptions and check robustness of MR results

Table 13.2 Selected Mendelian Randomization approaches based on GWAS summary statistics only or in combination with individual level data

Based on summary results						
Method	Short description	Software	Requirements	Main advantage	Main disadvantages	Paper
MR-Egger	• Using multiple individual genetic variants as instruments in separate models • Meta-analysing results • Adding an intercept indicating bias due to pleiotropy	https://raw.githubusercontent.com/remlap/mot/mrrobust/master/	GWAS summary statistics on X and Y for independent individuals	Models pleiotropy – relaxes exclusion restriction assumption	Assuming no correlation between genetic effects on X and pleiotropic effects (InSIDE) assumption)	Bowden et al. 2015 (38)
MR-Presso	• Using multiple individual genetic variants as instruments • Meta-analysing results • Detects and removes pleiotropic SNPs	https://github.com/rondolab/MR-PRESSO	Top SNPs from GWAS summary statistics on X and Y (ideally for independent individuals)	Outlier detection – improvement to MR-Egger	Requires InSIDE assumption, <50% of the SNPs are assumed to be pleiotropic	Verbanck et al. 2018 (40)
GSMR/Heidi	• Using multiple individual genetic variants as instruments • Removing pleiotropic SNPs • Meta-analysing results • Detects and removes pleiotropic SNPs • Takes LD into account	http://cnsgenomics.com/software/gsmr/	Top SNPs from GWAS summary statistics on X and Y (ideally for independent individuals) AND LD reference panel based on individual data	Outlier detection - more powerful than MR-Presso (taking LD into account)	Requires at least 10 SNPs with pvalue <5× 10 ⁻⁸	Zhu et al. 2018 (42)
Based on individual level data						
GIV	• Using PGS for X as IV • Controlling for PGS for Y conditional on X • Erasing measurement error for PGS Y	For score construction, e. g. PRSice, LDpred (see Chapter 10). For analysis, any data analysis software, e. g. R (see Appendix 1), Stata, SPSS.	Individual level genotype and phenotype data and GWAS summary statistics for X and Y excluding the prediction sample	Can control for environmental confounders or gene-environment correlation in regression framework.	Might be biased due to environmental confounders unrelated to genetic effects	DiPrete et al. 2018 (43)

Genetic pleiotropy

- One central assumption in Mendelian randomization is that the **genetic variants must not affect the outcome in any other way**. For example, there is a variant of the gene that encodes the enzyme aldehyde dehydrogenase (ALDH2) that disrupts metabolism of alcohol. When people with this variant drink, they tend to feel nauseous, and so it's associated with lower levels of alcohol consumption. **That might seem a plausible way to test, for example, whether drinking raises blood pressure, because those who carry the variant generally drink less than those who don't.**
- The problem is that ALDH2 also influences **how likely someone is to smoke**, which independently influences blood pressure. This phenomenon, known as **genetic pleiotropy**, can invalidate Mendelian randomization results. And that creates a problem, because the extent of pleiotropy isn't fully realized for many genes

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A 2 minutes Tutorial on the topic by George Davey Smith:

<https://www.youtube.com/watch?v=LoTgfGotaQ4>